

DDTA BA5 canonical semantic registry and controlled-authoring candidate - R2

Status: PROVISIONAL CANDIDATE AFTER BA5-T2 / BA5 OPEN

Repository baseline reviewed: 85622414b2ff52d58f3cd11776fef4b3753afc7d

Supersedes for active BA5 reasoning: the R1 canonical-referent naming candidate. The R1 artifact remains immutable BA5-T1 derivation history.

Closed dependencies: documentation layer; BA0 responsibility boundary; BA1 BAREferent + BAProposition; BA2 proposition/operator/role/classification contract; BA3 provenance/continuity; BA4 projection contract; BA5-T1 canonical referent naming lower bound.

1. Purpose

BA5-T2 asks whether the strict T1 authoring rule can cover the wider operative Base Analysis vocabulary without introducing a synonym ontology or letting a tool decide semantic equivalence.

The starting rule remains:

```
registered canonical token
    -> normative semantic authoring
```

```
unregistered alias / synonym
    -> not silently equivalent
```

T2 finds that the rule survives, but a single flat registry does not. Canonical tokens belong to semantic domains with different scoping, authority and extension rules.

2. Provisional R2 lower bound

The smallest current-scope registry model is a logical set of domain-scoped registries, not one universal token table:

```
CanonicalSemanticRegistrySet                                [logical umbrella; NOT BAE]
|
+-- CanonicalReferentNameRegistry                            [project-governed]
|   |- governedBaselineKey
|   |- namingScopeKey
|   `-- canonicalName -> governed referent binding
|
+-- SemanticOperatorRegistry                                [BA2 contract-governed]
|   |- registryRevisionKey immutable
|   `-- semanticOperatorKey -> normative BA2 meaning
|
+-- OperatorRoleRegistry                                    [BA2 contract-governed]
|   |- registryRevisionKey immutable
|   `-- (semanticOperatorKey, roleKey) -> BA2 role contract
|
+-- SemanticKindRegistry                                    [method-neutral, extensible]
|   |- registryRevisionKey immutable
|   `-- semanticKindKey -> normative method-neutral definition
|
`-- ControlledValueRegistry 0..*                             [domain-scoped, extensible where allowed]
    |- valueDomainKey
    |- registryRevisionKey immutable
    `-- canonicalValueKey -> normative domain meaning
```

This is a semantic responsibility model. It does not require five databases, five files, graph node kinds or ThreatForge classes. Physical co-location is allowed when domain, scope, authority and revision remain mechanically recoverable.

3. Why one flat namespace is rejected

Token equality across domains does not establish semantic equality.

For example, a project might legitimately contain a referent named `source` while BA2 also uses `source` as the role of `transfer`.

```
referent canonicalName = source
roleKey under transfer = source
```

Those tokens occupy different domains and must not collide or alias merely because their strings are identical.

Therefore uniqueness is domain-scoped:

```
same domain + same active scope + same canonical token
  -> one registry meaning/binding
```

```
same token in different domains
  -> allowed when each domain resolution is unambiguous
```

A global token uniqueness rule is **REJECTED AS OVER-STRONG**.

4. Referent names retain the T1 rule

For a named shared project referent:

```
one governed baseline / naming scope
  -> one exact canonicalName
```

The canonical name is project-governed lexical metadata, not BA identity. A governed rename may retain `BAReferent` identity through BA3 continuity.

Nothing in T2 weakens the T1 requirement that shared views use the same canonical referent name when presenting that shared entity.

5. Semantic operator registry coverage

All thirteen BA2 operator keys pass exact-token enforcement:

```
transfer
produce
create
observe
transition
correlate
reference
dependOn
consumeService
realize
assignResponsibility
constrain
classify
```

These tokens are not project-local synonyms. Their normative meaning belongs to the accepted BA2 methodology contract.

Examples:

```
send          -> not an accepted semanticOperatorKey
transfer      -> accepted canonical key
```

```
usesService   -> not an accepted semanticOperatorKey
consumeService -> accepted canonical key
```

Free explanatory prose may use natural language, but a machine-significant operator binding must resolve to the exact registered key. Hidden lexical mapping from `send` to `transfer` is not part of the BA5 core.

The operator registry is scoped to an immutable methodology/BA2 registry revision, not separately renamed by every project baseline.

6. Operator-scoped role registry

BA2 already rejects a context-free global role contract. T2 therefore rejects a flat role registry that stores `roleKey` without its operator context.

The canonical lookup key is conceptually:

```
(semanticOperatorKey, roleKey)
```

Examples from the closed BA2 contract:

`transfer:`

```
    source
    destination
    content
```

`produce:`

```
    actor
    result
    input
```

`create:`

```
    actor
    created
```

`observe:`

```
    actor
    observed
    result
```

`transition:`

```
    subject
    toState
    actor
    fromState
```

`correlate:`

```
    correlatedItem
    correlationContext
```

`reference:`

```
    referencer
    referenced
```

`dependOn:`

```
    dependent
    prerequisite
```

`consumeService:`

```
    consumer
    service
    provider
```

`realize:`

```
    abstract
    realization
```

`assignResponsibility:`

```
responsibleParty
responsibilityScope
responsibilityKind
```

```
constrain:
  constraintTarget
  constraintValue
```

```
classify:
  classifiedReferent
  semanticKind
```

The same lexical role token may recur under more than one operator without implying a context-free contract. Cardinality and required/optional semantics remain BA2 responsibilities; BA5 requires exact contextual lookup rather than redefining them.

Alias attempts such as `src`, `dest`, `owner` or `kind` do not silently normalize to registered roles.

7. Semantic-kind registry

BA2 closes a registry contract but intentionally does not close a universal semantic-kind taxonomy.

BA5-T2 therefore separates two questions:

```
Is semanticKind a controlled canonical domain?    YES
Is every possible semanticKind pre-enumerated?    NO
```

Current evidence gives clear exact examples such as:

```
store
contract
```

These can be used as canonical `semanticKind` tokens when their normative method-neutral definitions are registered.

BA2 also discusses slash-form conceptual examples:

```
party/person
information/resource
behavior/event
```

BA5-T2 does **not** silently turn this explanatory wording into canonical keys. Exact token admission must be explicit.

A useful-looking term such as:

```
channel
```

is therefore **CANDIDATE / EVIDENCE-GATED**, not automatically accepted merely because `Interface / Connection / Channel` was discussed during BA1 ontology pressure. If a governed corpus needs a method-neutral `channel` distinction that existing kinds cannot preserve, it may enter the governed extension workflow below.

8. Controlled-value registry domains

Not every non-referent value should become a registry token. T2 preserves the BA1/BA2 distinction between controlled vocabulary and ordinary typed/local values.

A controlled-value registry is justified when a value is selected from a reusable semantic domain whose distinctions must be stable across authoring and consumers.

Current BA2 examples include:

```
polarity:
  affirmative
  negative
```

```
scopedModifierKind:
  condition
  temporalScope
```

```
responsibilityKind:
  ownership
  management
... evidence-gated additional authority modes
```

For **polarity** and **scopedModifierKind**, the accepted BA2 contract already closes the exact current-scope values.

For **responsibilityKind**, BA2 requires a controlled domain but does not claim an exhaustive universal list. **ownership** and **management** are current evidence-backed values; new authority modes are admitted only through governance.

A literal such as a duration, number, URI, threshold or project-specific string does not become a canonical registry term merely because it appears in a proposition. If it needs independent reusable identity, BA1 promotes it to a **BAReferent**; otherwise it remains an appropriate typed/local value.

9. Registry revision identity

T2 finds that mutable registry meaning would break reproducibility even when the canonical token string stays unchanged.

Therefore any registry whose definitions/contracts can evolve requires immutable revision identity:

```
same token
+ registry revision R1
  may not be assumed equivalent to
same token
+ registry revision R2
```

A registry revision may preserve all prior meanings while adding a new entry, but the used revision remains inspectable.

The referent-name registry is already anchored by an immutable governed documentation baseline/naming scope. BA2 contract registries and extensible semantic/value registries require equivalent immutable revision resolution.

This refines registry governance; it does not create a new BAE lifecycle. BA3 already assumes reproducibility under the same accepted semantic registries.

10. Exact-token path across documentation, BA and projections

The accepted path is:

```
governed semantic authoring
-> exact domain-scoped canonical token
  -> accepted BA binding
    -> projection trace / shared rendering
```

For example:

```
referent: cameraIngresso
operator: transfer
role: transfer/source
kind: store
```

A human or method projection may use its own presentation/type vocabulary, but it must not rewrite the underlying shared key when claiming to expose shared BA semantics.

Allowed:

```
[FlowParticipant] cameraIngresso
method-owned kind = DataFlowEndpoint
BA trace -> transfer/source -> cameraIngresso
```

Rejected:

```
method label "sender"
  silently substitutes for BA roleKey source
```

Method-specific taxonomy remains projection-owned under BA4; it is not admitted into the BA registry merely because it is useful.

11. Governed extension workflow

When authoring needs an unregistered semantic token, the tool or author does not immediately add it.

The minimum decision path is:

```
unregistered semantic need
|
+-- lexical alias of existing registered meaning?
|   -> REJECT new entry
|   -> use existing canonical token
|
+-- method-specific/projection-local meaning?
|   -> keep downstream in projection/method registry
|
+-- independently reusable project-semantic identity?
|   -> represent as BAREferent, not controlled local value
|
+-- new entry inside an already extensible method-neutral domain?
|   -> candidate registry extension
|   -> evidence + normative definition + non-equivalence check
|   -> governed review
|   -> new immutable registry revision if accepted
|
`-- changes a closed BA2 operator/role/polarity/modifier contract?
    -> smallest BA2 reopen candidate
    -> BA5 cannot smuggle it in as lexical metadata
```

This is the central T2 authority boundary.

12. Minimum extension packet

For an extensible `semanticKind` or controlled-value domain, an accepted new canonical entry requires at least:

1. `registryDomain` / `valueDomainKey`;
2. proposed exact canonical token;
3. normative methodology-neutral definition;
4. concrete governed evidence or integration need;
5. comparison against existing entries showing that it is not merely an alias;
6. confirmation that the meaning is not method-specific;
7. confirmation that independent BAE identity is not the actual missing requirement;
8. governed acceptance; and
9. publication in an immutable registry revision used by subsequent authoring/BA baselines.

The tool may assemble and validate this packet. It may not approve it as semantic authority.

13. Closed BA2 contract versus extensible registry content

T2 makes an important distinction:

```
new semanticKind / controlled value
within an already open registry contract
  -> may be admitted by governed registry extension

new semanticOperatorKey
```

or changed operator-scoped role contract
or new polarity/modifier kind
 -> changes closed BA2 semantics
 -> smallest BA2 reopen required

This prevents BA5 from becoming a back door for modifying BA2.

A hypothetical new **publish** operator that is merely conveyance is rejected as an alias of **transfer**. If concrete evidence proves a distinct method-neutral action not representable by the thirteen accepted operators, that evidence belongs to BA2 reopen review, not an automatic vocabulary extension.

14. Tool-support boundary

A future tool may:

- validate exact tokens against the correct domain/revision;
- offer context-sensitive completion;
- after selecting **transfer**, offer only its accepted role keys;
- flag duplicate/colliding referent names inside the active naming scope;
- flag unregistered semantic kinds or controlled values;
- show normative definitions and registry provenance;
- classify an unregistered token as a candidate extension request;
- assemble evidence for human/governed review.

It may not:

- infer hidden synonym equivalence;
- pick a new canonical token and treat it as accepted;
- change an operator/role contract through a local project registry;
- promote method taxonomy into BA vocabulary;
- decide that a local value needs BAE identity; or
- mutate a registry definition in place.

ThreatForge remains an enforcement/integration instrument, never the semantic authority.

15. T2 mutation pressure

V1 - operator alias

`send -> transfer`

`send` is not accepted as a second normative operator token. Use **transfer** in the semantic binding.

Result: **PASS / ALIAS REJECTED.**

V2 - role alias and context loss

`transfer/src`

`src` is rejected. `source` is valid only through the **transfer** role contract. A context-free role lookup is insufficient.

Result: **PASS WITH DOMAIN/CONTEXT REFINEMENT.**

V3 - same token in different domains

`referent canonicalName = source`
`transfer roleKey = source`

No collision exists because the registries are domain-scoped.

Result: **PASS; FLAT GLOBAL NAMESPACE REJECTED.**

V4 - semantic-kind alias

repository

If the proposed definition is materially the same as registered **store**, reject the new token as a synonym. If governed evidence shows a distinct method-neutral concept, process it as a new semantic-kind candidate rather than guessing from the word.

Result: **PASS**.

V5 - candidate channel

channel is not admitted merely because it is a familiar systems term. No T2 evidence proves that current **BAReferent** + **classify** + **transfer/realize** semantics lose required shared meaning without it.

Result: **DEFERRED / EVIDENCE-GATED**.

V6 - responsibility kind

ownership and **management** remain controlled values. A proposed **control** value must receive a normative distinction and evidence; it cannot enter as a loose synonym.

Result: **PASS**.

V7 - genuinely new operator

A proposed operator whose meaning is not representable by the thirteen-key BA2 registry cannot be accepted through BA5 alone.

Result: **BA2 REOPEN PATH REQUIRED IF MATERIAL EVIDENCE SURVIVES**; no current reopen triggered.

V8 - projection terminology

A method may display **DataFlowEndpoint**, **TrustBoundary** or another method-owned type while tracing to canonical BA keys. Those labels do not enter the BA registry.

Result: **PASS**.

V9 - tool autocomplete

With **semanticOperatorKey** = **transfer**, the tool may offer **source**, **destination**, **content** from the immutable role contract. It may not offer **src** or invent another role based on usage frequency.

Result: **PASS**.

16. Reopen checks

No current T2 case forces a reopen of BA0-BA4.

- BA1: registry entries are not a third semantic identity family; independent identity still promotes to **BAReferent**.
- BA2: current operators/roles remain sufficient for the tested cases. The extension workflow explicitly routes a genuinely new operator/role back to BA2 rather than changing it silently.
- BA3: immutable registry revision resolution is compatible with its existing reproducibility assumption and does not add a project lifecycle.
- BA4: method-local taxonomy remains downstream; canonical BA token trace remains sufficient.
- Documentation metamodel: controlled authoring remains a cross-document governance constraint, not a new MR/Decision/FR/SR family.

17. Provisional dispositions

strict canonical-token authoring	REQUIRED
one flat global token namespace	REJECTED
registry domain/context resolution	REQUIRED
referent canonicalName baseline/naming scoped	REQUIRED

13 BA2 operator keys exact	REQUIRED
operator role lookup as (operatorKey, roleKey)	REQUIRED
role alias/synonym normalization	REJECTED
semanticKind as controlled extensible domain	REQUIRED
universal semanticKind taxonomy	REJECTED / NOT FORCED
store / contract as current exact semanticKind evidence	ACCEPTED FOR T2 PRESSURE
channel	DEFERRED / EVIDENCE-GATED
controlled reusable value domains	REQUIRED WHERE APPLICABLE
literal/local value -> registry entry by default	REJECTED
polarity affirmative negative	ACCEPTED EXACT KEYS
modifier kind condition temporalScope	ACCEPTED EXACT KEYS
responsibilityKind controlled domain	REQUIRED
ownership / management current values	ACCEPTED FOR T2 PRESSURE
immutable registry revision resolution	REQUIRED
alias candidate -> new registry entry	REJECTED
method-specific term -> BA registry	REJECTED
governed extensible-domain admission	REQUIRED
new operator/role through BA5 extension	REJECTED
material new operator/role -> smallest BA2 reopen	REQUIRED PATH
tool validation/context completion	ALLOWED
tool semantic approval/equivalence	REJECTED
new BAE family	NOT FORCED
BA1 / BA2 / BA3 / BA4 reopen	NOT TRIGGERED
BA5	STARTED / NOT CLOSED

18. Falsification rules

Revise this candidate if concrete corpus/integration evidence demonstrates that:

1. a normative alias must coexist with a canonical token in the same semantic domain to preserve necessary shared meaning;
2. role semantics cannot be resolved through operator-scoped canonical keys;
3. a flat registry is actually necessary for interoperability and domain separation loses required semantics;
4. a semantic-kind/controlled-value extension cannot be governed without turning the tool into authority;
5. immutable registry revisions are insufficient to replay the meaning used by an accepted BA baseline;
6. method projection semantics require method-specific taxonomy to enter the shared BA registry; or
7. the extension decision tree routes a genuine shared semantic need to the wrong layer.

Until such evidence exists, synonym/alias machinery remains unnecessary complexity.

19. Smallest unresolved set after BA5-T2

T1 and T2 now cover named referents, BA2 operators, operator-scoped roles, semantic kinds, controlled values, exact-token enforcement, extension governance and the tool authority boundary.

The smallest remaining BA5 task is an integrated closure review that tries to break the combined contract across:

- referent rename plus registry revision change;
- human and incompatible method projections;
- current facial/order examples;
- extensible versus closed registry domains;
- alias pressure; and
- tool completion/extension workflow.

If no material counterexample survives, BA5 may close for current thesis scope. Do not start BA6 before that closure is official.